

Contiuous Assement I

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Question 1

a The allowed range of the number of excited quanta distributed among N oscillators can be determined using the first equation

$$\epsilon_n^N = \sum_{i=1}^N r_i \hbar \omega = n \hbar \omega \quad (1)$$

Dividing by $\hbar \omega$ (ω is non-zero) gives us

$$\sum_{i=1}^N r_i = n \quad (2)$$

Since for any given $r \in \mathbb{Z}^+$ we have an infinite spectrum available thus we get the allowed values for n

$$n \in \{0, 1, 2, \dots\} (n \in \mathbb{Z}^+) \quad (3)$$

b

The lowest three energy levels possible correspond to the n values of $n = 0, 1, 2$ with energies $0, \hbar \omega$, and $2\hbar \omega$ respectively. Since there are 4 oscillators we have $N = 4$.

The expression for microstates given by

$$\Omega(N, n) = \frac{(N + n - 1)!}{(n)! (N - 1)!} \quad (4)$$

Thus we can compute the number of microstates for each value of n .

$$\Omega(4, 0) = \frac{(4 + 0 - 1)!}{(0)! (4 - 1)!} \quad (5)$$

$$\Omega(4, 0) = 1 \quad (6)$$

So there is only 1 possible microstate for the ground state.

$$\Omega(4, 1) = \frac{(4 + 1 - 1)!}{(1)! (4 - 1)!} \quad (7)$$

$$\Omega(4, 1) = 4 \quad (8)$$

So there are 4 microstates for the first energy level $n = 1$.

$$\Omega(4, 2) = \frac{(4 + 2 - 1)!}{(2)! (4 - 1)!} \quad (9)$$

$$\Omega(4, 2) = 10 \quad (10)$$

XXX Include figures

c

We have an expression for entropy give by

$$S = k \ln(\Omega) \quad (11)$$

Plugging in our expression for the number of micro-states we get

$$S = k \ln \left(\frac{(N + n - 1)!}{(n)! (N - 1)!} \right) \quad (12)$$

We can use the laws of logs with $\ln \left(\frac{a}{b} \right) = \ln(a) - \ln(b)$ and $\ln(\alpha\beta) = \ln(\alpha) + \ln(\beta)$. We obtain

$$S = k \ln((N + n - 1)!) - \ln(n!) - \ln((N - 1)!) \quad (13)$$

$$S = k \ln((N + n - 1)!) - \ln(n!) - \ln((N - 1)!) \quad (14)$$

Now we can use the Stirlings Approximation which says that

$$\ln(\rho!) \approx \rho \ln(\rho) - \rho \quad (15)$$

We can apply it three times above to receive

$$\frac{S}{k} = [(N+n-1) \ln(N+n-1) - (N+n-1)] - [n \ln(n) - n] - [(N-1) \ln(N-1) - (N-1)] \quad (16)$$

We can drop the terms of order 1 as it is safe to assume that $N \gg 1$. Thus we get

$$\frac{S}{k} = [(N+n) \ln(N+n) - (N+n)] - [n \ln(n) - n] - [(N) \ln(N) - (N)] \quad (17)$$

We can expand to get

$$\frac{S}{k} = [N \ln(N+n) + n \ln(N+n) - N - n - n \ln(n) + n - N \ln(N) + N] \quad (18)$$

We can take out a factor of N inside the logarithms and also cancel terms to get

$$\frac{S}{k} = N \ln\left(N\left(1 + \frac{n}{N}\right)\right) + n \ln\left(N\left(1 + \frac{n}{N}\right)\right) - n \ln(n) - N \ln(N) \quad (19)$$

Using the product law of logs we get

$$\frac{S}{k} = N \ln(N) + N \ln\left(1 + \frac{n}{N}\right) + n \ln N + n \ln\left(1 + \frac{n}{N}\right) - n \ln n - N \ln N \quad (20)$$

$$\frac{S}{k} = N \ln\left(1 + \frac{n}{N}\right) + n \ln N + n \ln\left(1 + \frac{n}{N}\right) - n \ln n \quad (21)$$

$$\frac{S}{kN} = \ln\left(1 + \frac{n}{N}\right) + \frac{n}{N} \ln N + \frac{n}{N} \ln\left(1 + \frac{n}{N}\right) - \frac{n}{N} \ln n \quad (22)$$

$$\frac{S}{kN} = \left(1 + \frac{n}{N}\right) \ln\left(1 + \frac{n}{N}\right) + \frac{n}{N} \ln N - \frac{n}{N} \ln n \quad (23)$$

$$\frac{S}{kN} = \left(1 + \frac{n}{N}\right) \ln\left(1 + \frac{n}{N}\right) - \frac{n}{N} \ln \frac{n}{N} \quad (24)$$

We obtain the required form of entropy

$$S = kN \left[\left(1 + \frac{n}{N}\right) \ln\left(1 + \frac{n}{N}\right) - \frac{n}{N} \ln\left(\frac{n}{N}\right) \right] \quad (25)$$

d

Given that $U = n\hbar\omega$.

We can substitute this into (??) above to obtain

$$S = kN \left[\left(1 + \frac{U}{N\hbar\omega}\right) \ln\left(1 + \frac{U}{N\hbar\omega}\right) - \frac{U}{N\hbar\omega} \ln\left(\frac{U}{N\hbar\omega}\right) \right] \quad (26)$$

e

To find the energy as a function of temperature $U(T)$ we have

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{V,N} \quad (27)$$

$$\frac{1}{TkN} = \left(\frac{1}{N\hbar\omega} \right) \ln\left(1 + \frac{U}{N\hbar\omega}\right) + \left(1 + \frac{U}{N\hbar\omega}\right) \left(1 + \frac{U}{N\hbar\omega}\right)^{-1} \left(\frac{1}{N\hbar\omega} \right) - \left(\frac{1}{N\hbar\omega} \right) \ln\left(\frac{U}{N\hbar\omega}\right) - \left(\frac{U}{N\hbar\omega} \right) \left(\frac{U}{N\hbar\omega} \right)^{-1} \left(\frac{1}{N\hbar\omega} \right) \quad (28)$$

Cancelling terms

$$\frac{1}{TkN} = \left(\frac{1}{N\hbar\omega} \right) \ln \left(1 + \frac{U}{N\hbar\omega} \right) - \left(\frac{1}{N\hbar\omega} \right) \ln \left(\frac{U}{N\hbar\omega} \right) \quad (29)$$

Rearranging

$$\frac{\hbar\omega}{Tk} = \ln \left(1 + \frac{U}{N\hbar\omega} \right) - \ln \left(\frac{U}{N\hbar\omega} \right) \quad (30)$$

$$\frac{\hbar\omega}{Tk} = \ln \left(\frac{N\hbar\omega}{U} + 1 \right) \quad (31)$$

exponentiating both sides

$$e^{\frac{\hbar\omega}{Tk}} = \frac{N\hbar\omega}{U} + 1 \quad (32)$$

reordering

$$(N\hbar\omega)^{-1} \left(e^{\frac{\hbar\omega}{Tk}} - 1 \right) = \frac{1}{U} \quad (33)$$

Flipping both sides we obtain the required result

$$U = \frac{N\hbar\omega}{e^{\frac{\hbar\omega}{Tk}} - 1} \quad (34)$$

Question 2

a

We have the definition of the partition function in the grand canonical ensemble

$$\Xi = \sum_{N=0}^{\infty} \sum_{i_N} e^{\left(\frac{N\mu - E_{i_N}}{kT} \right)} \quad (35)$$

We are given energy constraints of $E_0 = 0$, $E_1 = 0$ and $E_2 > 0$ Thus we obtain

$$\Xi = e^{\frac{0}{kT}} + e^{\frac{\mu}{kT}} + e^{\frac{\mu - E_2}{kT}} \quad (36)$$

$$\Xi = 1 + e^{\frac{\mu}{kT}} + e^{\frac{\mu - E_2}{kT}} \quad (37)$$

b

By definition we have

$$\langle N \rangle = kt \frac{\partial \ln \Xi}{\partial \mu} = \frac{kT}{\Xi} \frac{\partial \Xi}{\partial \mu} \quad (38)$$

We can sub in our systems partition function

$$\langle N \rangle = \frac{kT}{1 + e^{\frac{\mu}{kT}} + e^{\frac{\mu - E_2}{kT}}} \left(0 + \frac{e^{\frac{\mu}{kT}}}{kT} + \frac{e^{\frac{\mu - E_2}{kT}}}{kT} \right) \quad (39)$$

Simplifying

$$\langle N \rangle = \frac{e^{\frac{\mu}{kT}} + e^{\frac{\mu - E_2}{kT}}}{1 + e^{\frac{\mu}{kT}} + e^{\frac{\mu - E_2}{kT}}} \quad (40)$$

Which can equivalently be expressed solely in terms of the partition function as

$$\langle N \rangle = \frac{\Xi - 1}{\Xi} \quad (41)$$

c

Previously we had the thermal average occupancy for all energies. However, now we are only concerned with the energy E_2

$$\langle N(E_2) \rangle = \frac{e^{\frac{\mu - E_2}{kT}}}{1 + e^{\frac{\mu}{kT}} + e^{\frac{\mu - E_2}{kT}}} = \frac{e^{\frac{\mu - E_2}{kT}}}{\Xi} \quad (42)$$

d

Using the definition of the average of a thermodynamic potential we have

$$\langle E_i \rangle = \frac{1}{\Xi} \sum_{N=0}^{\infty} \sum_{i_N} E_i e^{\frac{N\mu - E_i}{kT}} \quad (43)$$

$$\langle E_2 \rangle = \frac{kT}{1 + e^{\frac{\mu}{kT}} + e^{\frac{\mu - E_2}{kT}}} \left(0 + E_1 \frac{e^{\frac{\mu}{kT}}}{kT} + E_2 \frac{e^{\frac{\mu - E_2}{kT}}}{kT} \right) \quad (44)$$

But $E_1 = 0$

$$\langle E_2 \rangle = \frac{kT}{1 + e^{\frac{\mu}{kT}} + e^{\frac{\mu - E_2}{kT}}} \left(E_2 \frac{e^{\frac{\mu - E_2}{kT}}}{kT} \right) = \langle E_2 \rangle = E_2 \langle N(E_2) \rangle \quad (45)$$

e

$$\Xi_{\text{Sim}} = \sum_{N=0}^{\infty} \sum_{i_N} e^{\left(\frac{N\mu - E_{i_N}}{kT} \right)} \quad (46)$$

$$\Xi_{\text{Sim}} = e^{\frac{0}{kT}} + e^{\frac{\mu}{kT}} + e^{\frac{\mu - E_2}{kT}} + e^{\frac{2\mu - (E_1 + E_2)}{kT}} \quad (47)$$

$$\Xi_{\text{Sim}} = 1 + e^{\frac{\mu}{kT}} + e^{\frac{\mu - E_2}{kT}} + e^{\frac{2\mu - (E_1 + E_2)}{kT}} \quad (48)$$

Question 3

a

The Hamiltonian of an ideal gas is given by

$$H = \frac{p^2}{2m} \quad (49)$$

We can take the ansatz for the wavefunction to be

$$\psi = Ae^{ax} + Be^{-bx} \quad (50)$$

into

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi = E\psi \quad (51)$$

Using boundary conditions that $\psi(0) = \psi(L) = 0$ and $\int_0^L |\psi(x)|^2 dx = 1$ We obtain the standing wave solutions to a 1-D potential well.

$$\psi = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad (52)$$

with $n = 1, 2, 3, \dots$

Since our hamiltonian is independent of time we can say that $H = E$ and we know

$$k = \sqrt{\frac{2mE}{\hbar^2}} \quad (53)$$

Which gives us the energy of the n th level

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad (54)$$

b

We are given in lectures that

$$N = \sum_N f^{MB}(\epsilon_n) = \sum_{n=1}^{\infty} e^{\frac{\mu - \epsilon_n}{kT}} \quad (55)$$

We can take the term independent of the summation index out

$$N = e^{\frac{\mu}{kT}} \sum_{n=1}^{\infty} e^{-\epsilon_n} \quad (56)$$

Solving for our chemical potential μ (hopefully our sum isn't zero)

$$\mu = kT \ln \left[\frac{N}{\sum_{n=1}^{\infty} e^{-\epsilon_n}} \right] \quad (57)$$

Now we can examine our summation $\clubsuit$ explicitly

$$\clubsuit = \sum_{n=1}^{\infty} e^{-\epsilon_n} = \sum_{n=1}^{\infty} e^{-\frac{\pi^2 \hbar^2}{2mL^2}} \quad (58)$$

In this regime we can approximate the summation as an integral as we have $\Delta\epsilon_n \ll kT$ for the standard masses and lengths we are interested in. We can also perform a substitution for convince of $\alpha = \frac{\hbar^2 \pi^2}{2mL^2 kT}$

$$\clubsuit = \int_0^{\infty} e^{-an^2} \quad (59)$$

We can notice that this is very similar to a standard gaussian integral except for the standard gaussian integral we have an integral from $-\infty$ to ∞ . However, if we examine our integrand we can see that it is an even function thus we can extend our integration bounds while at the same time introducing a factor of $\frac{1}{2}$.

$$\clubsuit = \frac{1}{2} \int_{-\infty}^{\infty} e^{-an^2} \quad (60)$$

Now we see that the Gaussian integral can be identified. A standard Gaussian integral is below:

$$\int_{-\infty}^{\infty} x^{2n} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}} \frac{(2n-1)!!}{(2\alpha)^n} \quad (61)$$

In our case we have $n = 0$ Thus we get

$$\clubsuit = \frac{1}{2} \sqrt{\frac{\pi}{\alpha}} = \frac{1}{2} \sqrt{\frac{2mL^2 kT}{\hbar^2 \pi}} \quad (62)$$

If we bring the $\frac{1}{2}$ inside we can identify the quantum concentration in this case as $n_Q = \sqrt{\frac{mkT}{2\hbar^2 \pi}}$. Thus we get an expression for the chemical potential to be

$$\mu = kT \ln \left[\frac{N}{n_Q L} \right] \quad (63)$$

One could also label the linear concentration as $n_L = \frac{N}{L}$. So we arrive at our final expression

$$\mu = kT \ln \left[\frac{n_L}{n_Q} \right] \quad (64)$$

c

To find the internal energy we can use the fact that we have a large number of particles ie $N \gg 1$ thus we can use the approximation that

$$U \approx \langle \epsilon \rangle \quad (65)$$

Since we are working in the 1-D Maxwell Distribution we know

$$U = \sum_n \epsilon_n f^{MB}(\epsilon_n) \quad (66)$$

Using the definition of f^{MB} and ϵ_n

$$U = \sum_n \frac{\hbar^2 \pi^2}{2mL^2} e^{\frac{\mu}{kT}} e^{-\frac{\hbar^2 \pi^2}{2mL^2 kT}} \quad (67)$$

We see we have a sum which only depends on n so we can take non n dependent terms outside the sum and making the sub $a = \frac{\hbar^2 \pi^2}{2mL^2 kT}$

$$\frac{\hbar^2 \pi^2}{2mL^2} e^{\frac{\mu}{kT}} \sum_{n=0}^{\infty} n^2 e^{-an^2} \quad (68)$$

We can now perform our approximation as the sum becomes an integral letting $e^{\frac{\mu}{kT}} = \frac{N}{Ln_Q}$.

$$U = \frac{\hbar^2 \pi^2 N}{2mL^3 n_Q} \int_0^{\infty} n^2 e^{-an^2} \quad (69)$$

we recognise the gaussian intergral as seen in eqn (??) in this case $n = 1$

$$\int_0^{\infty} x^2 e^{-ax^2} dx = \frac{\sqrt{\pi}}{4a^{\frac{3}{2}}} \quad (70)$$

Thus we have

$$U = \frac{\hbar^2 \pi^2 N}{2mL^3} \frac{1}{n_Q} \frac{\sqrt{\pi}}{4a^{3/2}} \quad (71)$$

In expanded form this is

$$U = \frac{\hbar^2 \pi^2 N}{2mL^3} \left(\frac{mkT}{2\hbar^2 \pi} \right)^{-\frac{1}{2}} \left(\frac{\hbar^2 \pi^2}{2mL^2 kT} \right)^{-\frac{3}{2}} \frac{\sqrt{\pi}}{4} \quad (72)$$

We have the internal energy of a 1-D system given by

$$U = \frac{1}{2} NkT \quad (73)$$

Using the equipartition theorem we have for a D dimensional gas the internal energy is given by

$$U = \frac{D}{2} NkT \quad (74)$$

In three dimensions we have

$$U = \frac{3}{2} NkT \quad (75)$$

In the 1-D case there is only one translational degree of freedom. The corresponding translational energy is that of a 1d free particle $\frac{p^2}{2m}$. IN the three dimensional case there are three translational degrees of freedom and thus the translational energy of each atom is $\frac{(p_x + p_y + p_z)^2}{2m}$.

To find the entropy first we could find the Helmholtz Free Energy.

$$\mu = \left(\frac{\partial F}{\partial N} \right)_{T,L} \implies F = \int_0^N \mu(N) dN \quad (76)$$

$$F = \int_0^N kT \ln \left(\frac{N}{Ln_Q} \right) dN = kT \int_0^N [\ln(N) - \ln(Ln_Q)] dN \quad (77)$$

using

$$\int \ln x dx = x \ln x - x \quad (78)$$

we get

$$F = kTN \left(\ln \left(\frac{N}{L} - 1 - \ln(n_Q) \right) \right) \quad (79)$$

$$F = kTN \left(\ln \left(\frac{N}{L} - 1 - \ln \left(\frac{mkT}{2\hbar^2 \pi} \right)^{\frac{1}{2}} \right) \right) \quad (80)$$

$$F = kTN \left(\ln \left(\frac{N}{L} - 1 - \ln \left(\frac{mk}{2\hbar^2\pi} \right)^{\frac{1}{2}} \right) - \frac{1}{2} \ln T \right) \quad (81)$$

We have

$$\mathcal{S} = - \left(\frac{\partial F}{\partial T} \right)_{L,N} \quad (82)$$

$$\mathcal{S} = -kN \left(\ln \left(\frac{N}{L} \right) - 1 - \ln \left(\frac{mk}{2\hbar^2\pi} \right)^{\frac{1}{2}} - \frac{1}{2} \ln T \right) + \frac{kN}{2} \quad (83)$$

We can write this in a simpler form in terms of n_Q

$$\mathcal{S} = -kN \ln \left(\left(\frac{N}{Ln_Q} \right) - \frac{3}{2} \right) \quad (84)$$

Question 4

a

We have the energy levels with $r = 0, 1, 2, \dots$

$$E_r = \left(r + \frac{1}{2} \right) \hbar\omega \quad (85)$$

Here we consider the discrete partition function as we have quantised energy levels.

$$\mathcal{Z}_1 = \sum_{r=0}^{\infty} e^{-\beta E_r} \quad (86)$$

$$\mathcal{Z}_1 = \sum_{r=0}^{\infty} e^{-\beta(r+\frac{1}{2})\hbar\omega} \quad (87)$$

$$\mathcal{Z}_1 = e^{-\frac{1}{2}\beta\hbar\omega} \sum_{r=0}^{\infty} e^{-\beta r\hbar\omega} \quad (88)$$

Using the standard geometric series with $x = e^{-\beta\hbar\omega}$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots \quad (89)$$

We get

$$\mathcal{Z}_1 = \frac{e^{-\frac{1}{2}\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} \quad (90)$$

We can notice that this is similar to $\sinh x = \frac{e^x - e^{-x}}{2}$ so $\operatorname{csch} x = \frac{2}{e^x - e^{-x}}$. Thus we have an equivalent expression.

$$\mathcal{Z}_1 = \frac{1}{2} \operatorname{csch} \left(\frac{\hbar\omega\beta}{2} \right) \quad (91)$$

In terms of T

$$\mathcal{Z}_1 = \frac{1}{2} \operatorname{csch} \left(\frac{\hbar\omega}{2kT} \right) \quad (92)$$

Here we describe a QM harmonic oscillator in thermal equilibrium. This system is the vibrational modes of atoms in a lattice for a solid. We can see that each atom's behavior would be independent of its coordinates (position in the lattice) and they vibrate about the equilibrium position. (see part **b**)

b

To find the Helmholtz Free Energy for the single oscillator we use

$$\mathcal{F}_1 = -kT \ln \mathcal{Z}_1 \quad (93)$$

$$\mathcal{F}_1 = -kT \ln \frac{e^{-\frac{1}{2}\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} \quad (94)$$

$$\mathcal{F}_1 = -kT \left[-\frac{1}{2}\beta\hbar\omega - \ln(1 - e^{-\beta\hbar\omega}) \right] \quad (95)$$

In the natural variables (multiplying in) note there is no explicit N or V dependence

$$\mathcal{F}_1(T, V, N) = \frac{\hbar\omega}{2} + kT \ln(1 - e^{-\frac{\hbar\omega}{kT}}) \quad (96)$$

C

We can use the following for the mean energy

$$\langle E \rangle = -\frac{\partial \ln \mathcal{Z}_1}{\partial \beta} \quad (97)$$

$$\langle E \rangle = -\frac{\partial}{\partial \beta} \left[-\frac{1}{2}\beta\hbar\omega - \ln(1 - e^{-\beta\hbar\omega}) \right] \quad (98)$$

To extend our hyperbolic description we can identify that $\coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}} = \frac{1 + e^{-2x}}{1 - e^{-2x}}$ which can be seen if we cosine the exponential form below:

$$\langle E \rangle = \left[+\frac{1}{2}\hbar\omega + \frac{\hbar\omega e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} \right] = \frac{\hbar\omega}{2} \coth\left(\frac{\hbar\omega\beta}{2}\right) \quad (99)$$

The total internal energy can be found using

$$U = 3N \langle E \rangle \quad (100)$$

$$U = 3N \left[\frac{1}{2}\hbar\omega + \frac{\hbar\omega e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} \right] = \frac{3N\hbar\omega}{2} \coth\left(\frac{\hbar\omega\beta}{2}\right) \quad (101)$$

Here we see if we take the limit as $T \rightarrow 0$ only the first term survives. Thus the first term is known as the zero-point energy which means that there will be vibrational motion even at absolute zero, unlike the classical case.

D

In class we had the following expression for the Bose-Einstein Distribution

$$C_V^{phonon} = 3Nk \left(\frac{\hbar\omega}{kT} \right)^2 \frac{e^{\frac{\hbar\omega}{kT}}}{\left(e^{\frac{\hbar\omega}{kT}} - 1 \right)^2} \quad (102)$$

The heat capacity at constant volume is given by

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V \quad (103)$$

Using chain rule we have

$$C_V = 3N \frac{\partial \beta}{\partial T} \frac{\partial}{\partial \beta} \left[\frac{1}{2}\hbar\omega - \frac{\hbar\omega e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} \right] \quad (104)$$

Before computing the derivative multiplying top and bottom by $e^{\beta\hbar\omega}$ makes life worth living

$$C_V = - \left(-\frac{3N}{kT^2} \right) \left[\frac{\hbar\omega}{e^{\beta\hbar\omega} - 1} \right]^2 e^{\beta\hbar\omega} \quad (105)$$

Writing in terms of T

$$C_V = 3Nk \left(\frac{\hbar\omega}{kT} \right)^2 \frac{e^{\frac{\hbar\omega}{kT}}}{\left(e^{\frac{\hbar\omega}{kT}} - 1 \right)^2} \quad (106)$$

We can use the substitution $\theta_E = \frac{\hbar\omega}{k}$

$$C_V = 3Nk \left(\frac{\theta_E}{T} \right)^2 \frac{e^{\frac{\theta_E}{T}}}{\left(e^{\frac{\theta_E}{T}} - 1 \right)^2} \quad (107)$$

We notice the first term is the exact same as the result from class without a degeneracy factor of 3. This factor of 3 arises due to the three vibrational spacial degrees of freedom. The Einstein model assumes only one vibrational mode.

We can also examine the high temperature limit where we take $T \rightarrow \infty$ and we obtain $C_v \approx Nk$. In the low temperature limit $T \rightarrow 0$ we get $C_V \approx Nk \left(\frac{\hbar\omega}{kT} \right)^2 e^{-\frac{\hbar\omega}{kT}}$ if we were to taylor expand our exponential the larger power terms would dominate but as this is to a negative power C_V would vanish.

$$\mathcal{S} = k \ln \Omega \quad (108)$$

we have the multiplicity function with $3N$ oscillators

$$\Omega = \frac{(3N + n - 1)!}{n! (3N - 1)!} \quad (109)$$

$$\mathcal{S} = k \ln \left(\frac{(3N + n - 1)!}{n! (3N - 1)!} \right) \quad (110)$$

Using Stirlings approximation we get

$$\mathcal{S} \approx 3kN \left[\left(1 + \frac{n}{3N} \right) \ln \left(1 + \frac{n}{3N} \right) - \frac{n}{3N} \ln \left(\frac{n}{3N} \right) \right] \quad (111)$$

$$U = 3N\hbar\omega \left(\frac{1}{2} + \langle r \rangle \right) \quad (112)$$

where $\langle r \rangle$ is the average quanta of energy. But we know the total quanta is $n = 3N\langle r \rangle$ So we can substitute for $\langle r \rangle$

$$U = 3N\hbar\omega \left(\frac{1}{2} + \frac{n}{3N} \right) \quad (113)$$

We also know from part c that $U = 3N\langle E \rangle$ and we have

$$U = 3N\hbar\omega \left(\frac{1}{2} + \frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \right) \quad (114)$$

If we let these relations for U equal each other eq

$$\frac{n}{3N} = \frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \quad (115)$$

We can plug this into our expression for entropy to get

$$\mathcal{S} = 3kN \left[\left(1 + \frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \right) \ln \left(1 + \frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \right) - \frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \ln \left(\frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \right) \right] \quad (116)$$

$$\mathcal{S} = 3kN \left[\frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \left(\ln \left(1 + \frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \right) - \ln \left(\frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \right) \right) + \ln \left(1 + \frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \right) \right] \quad (117)$$

$$\mathcal{S} = 3kN \left[\frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \left(\ln \left(e^{\frac{\hbar\omega}{kT}} \right) \right) + \ln \left(1 + \frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \right) \right] \quad (118)$$

Thus we get an expression for the entropy to be

$$\mathcal{S}(T, N) = 3kN \left[\frac{\hbar\omega}{kT \left(e^{\frac{\hbar\omega}{kT}} - 1 \right)} + \ln \left(1 + \frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \right) \right] \quad (119)$$

f

We can take limits as $T \rightarrow 0$.

Notice

$$\ln \left(e^{\frac{\hbar\omega}{kT}} - 1 \right) \tag{120}$$

Thus the first term turns into $\ln 1$ which vanishes so the first term vanishes.

For the second term we find the following fact useful

$$\frac{\hbar\omega e^{-\frac{\hbar\omega}{kT}}}{kT \left(1 - e^{-\frac{\hbar\omega}{kT}} \right)} = \frac{\hbar\omega}{kT e^{\frac{\hbar\omega}{kT}} - kT} \tag{121}$$

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g

We are told that our answer should be in terms of the quantum concentration which is given by

$$n_Q = \left(\frac{2\pi m k T}{h^2} \right)^{\frac{1}{2}} \quad (122)$$

The Fermi-Dirac Partition function was given in class as XXX SLIDE 71 DO INTEGRAL

$$\mathcal{Z}^{FD} = \sum_{n=i}^1 e^{\frac{n_i(\mu - \epsilon_i)}{kT}} = 1 + e^{\frac{\mu - \epsilon_i}{kT}} \quad (123)$$

Since we are told that the ideal gas is at a fixed temperature T with a fixed number of particles N . We can use the canonical ensemble which gives a partition function for a single particle

$$\mathcal{Z}_1 = \int_0^\infty g(E) e^{-\beta E} dE \quad (124)$$

XXX INTEGRAL 0 to inf and introduce 2 where $\beta = \frac{1}{kT}$ and $g(E)$ is the density of states. For this system we have

$$\mathcal{Z}_1 = \int_0^\infty \frac{L}{h} e^{-\beta \frac{p^2}{2m}} \quad (125)$$

This is a standard gaussian integral

$$I = \int_{-\infty}^\infty e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}} \quad (126)$$

Thus we get

$$\mathcal{Z}_1 = \frac{L}{h} \sqrt{\frac{2\pi m}{\beta}} \quad (127)$$

Since we have N indistinguishable particles we raise an individual partition function to the power of N . We also introduce the $\frac{1}{N!}$ factor in order to avoid overcounting.

$$\mathcal{Z}_N = \frac{1}{N!} \mathcal{Z}_1^N \quad (128)$$

So the partition function for our system of N particles is

$$\mathcal{Z}_N = \frac{1}{N!} \frac{L^N}{h^N} \left[\frac{2\pi m}{\beta} \right]^{\frac{N}{2}} \quad (129)$$

One could use the relation for the chemical potential μ

$$\mu = -T \left(\frac{\partial S}{\partial N} \right)_{V,U} \quad (130)$$

So we want to find the entropy S which can be found from the Helmholtz free energy so we shall find the Helmholtz free energy which is given by

$$\mathcal{F} = -kT \ln \mathcal{Z}_N \quad (131)$$

Then we can use

$$S = - \left(\frac{\partial \mathcal{F}}{\partial T} \right)_{V,N} \quad (132)$$

So if we plug our partition function into (??)

$$- \frac{\mathcal{F}}{kT} = \ln \left[\frac{1}{N!} \frac{L^N}{h^N} \left[\frac{2\pi m}{\beta} \right]^{\frac{N}{2}} \right] \quad (133)$$

$$-\frac{\mathcal{F}}{hT} = \ln\left(\frac{1}{N!}\right) + \ln\left(\left(\frac{L}{h}\right)^N\right) + \ln\left(\left(\frac{2\pi m}{\beta}\right)^{\frac{N}{2}}\right) \quad (134)$$

$$-\frac{\mathcal{F}}{hT} = N \ln\left(\frac{L}{h}\right) - \ln(N!) + \frac{N}{2} \ln(2\pi m) - \frac{N}{2} \ln(\beta) \quad (135)$$

$$\mathcal{F} = -hT \left[N \ln\left(\frac{L}{h}\right) - \ln(N!) + \frac{N}{2} \ln(2\pi m) - \frac{N}{2} \ln(\beta) \right] \quad (136)$$

$$\mathcal{F} = -hT \left[N \ln\left(\frac{L}{h}\right) - \ln(N!) + \frac{N}{2} \ln(2\pi m) + \frac{N}{2} \ln(k) + \frac{N}{2} \ln(T) \right] \quad (137)$$

$$-\left(\frac{\partial \mathcal{F}}{\partial T}\right)_{V,N} = h \left[N \ln\left(\frac{L}{h}\right) - \ln(N!) + \frac{N}{2} \ln(2\pi m) + \frac{N}{2} \ln(k) + \frac{N}{2} \ln(T) \right] + hT \left[N \ln\left(\frac{L}{h}\right) - \ln(N!) + \frac{N}{2} \ln(2\pi m) + \frac{N}{2} \ln(k) + \frac{N}{2} \ln(T) \right] \quad (138)$$

$$-\left(\frac{\partial \mathcal{F}}{\partial T}\right)_{V,N} = h \left[(1+T) N \ln\left(\frac{L}{h}\right) - (T+1) \ln(N!) + (1+T) \frac{N}{2} (\ln(2\pi m) + \ln(k)) + \frac{N}{2} \ln T + \frac{N}{2} \right] \quad (139)$$

We can use the Stirlings approximation to get

$$-\left(\frac{\partial \mathcal{F}}{\partial T}\right)_{V,N} = h \left[(1+T) N \ln\left(\frac{L}{h}\right) - (T+1) (N \ln N - N) + (1+T) \frac{N}{2} (\ln(2\pi m) + \ln(k)) + \frac{N}{2} \ln T + \frac{N}{2} \right] \quad (140)$$

So our expression for the entropy is

$$\mathcal{S} = h \left[(1+T) N \ln\left(\frac{L}{h}\right) - (T+1) (N \ln N - N) + (1+T) \frac{N}{2} (\ln(2\pi m) + \ln(k)) + \frac{N}{2} \ln T + \frac{N}{2} \right] \quad (141)$$

We can find the chemical potential using (??)

$$\left(\frac{\partial \mathcal{S}}{\partial N}\right)_{V,U} = h \left[(1+T) \ln\left(\frac{L}{h}\right) - (1+T) (\ln N) + (1+T) \frac{1}{2} (\ln(2\pi m) + \ln k) + \frac{1}{2} \ln T + \frac{1}{2} \right] \quad (142)$$

$$\mu = -Th \left[(1+T) \ln\left(\frac{L}{h}\right) - (1+T) (\ln N) + (1+T) \frac{1}{2} (\ln(2\pi m) + \ln k) + \frac{1}{2} \ln T + \frac{1}{2} \right] \quad (143)$$

$$\mu = -Th \left[(1+T) \ln\left(\frac{L}{h}\right) - (1+T) (\ln N) + (1+T) \frac{1}{2} (\ln(2\pi mk)) + \frac{1}{2} \ln T + \frac{1}{2} \right] \quad (144)$$

$$\mu = -Th \left[\ln\left(\frac{L}{h}\right) + T \ln\left(\frac{L}{h}\right) - \ln N - T \ln N + \ln\left((2\pi mk)^{\frac{1}{2}}\right) + T \ln\left((2\pi mk)^{\frac{1}{2}}\right) + \ln T^{\frac{1}{2}} + \frac{1}{2} \right] \quad (145)$$

$$\mu = -Th \left[\ln\left(\frac{L}{h}\right) + T \ln\left(\frac{L}{h}\right) - \ln N - T \ln N + \ln\left((2\pi mkT)^{\frac{1}{2}}\right) + T \ln\left((2\pi mk)^{\frac{1}{2}}\right) + \frac{1}{2} \right] \quad (146)$$

$$\mu = -Th \left[\ln\left((2\pi mkT)^{\frac{1}{2}} \frac{L}{Nh}\right) + T \ln\left((2\pi mk)^{\frac{1}{2}} \frac{L}{Nh}\right) + \frac{1}{2} \right] \quad (147)$$

$$\mu = -Th \left[(1+T) \ln\left(\left(\frac{2\pi mkT}{h^2}\right)^{\frac{1}{2}} \frac{L}{N}\right) + \frac{1}{2} \right] \quad (148)$$

$$\mu = -Th \left[(1+T) \ln\left((n_Q)^{\frac{1}{3}} \frac{L}{N}\right) + \frac{1}{2} \right] \quad (149)$$

but the definition of linear concentration is $n_L = \frac{N}{L}$. Thus in terms of linear concentration and quantum concentration we have

$$\mu = \frac{-Th}{3} \left[(1+T) \ln\left(\left(\frac{n_Q}{n_L^3}\right)\right) + \frac{3}{2} \right] \quad (150)$$